



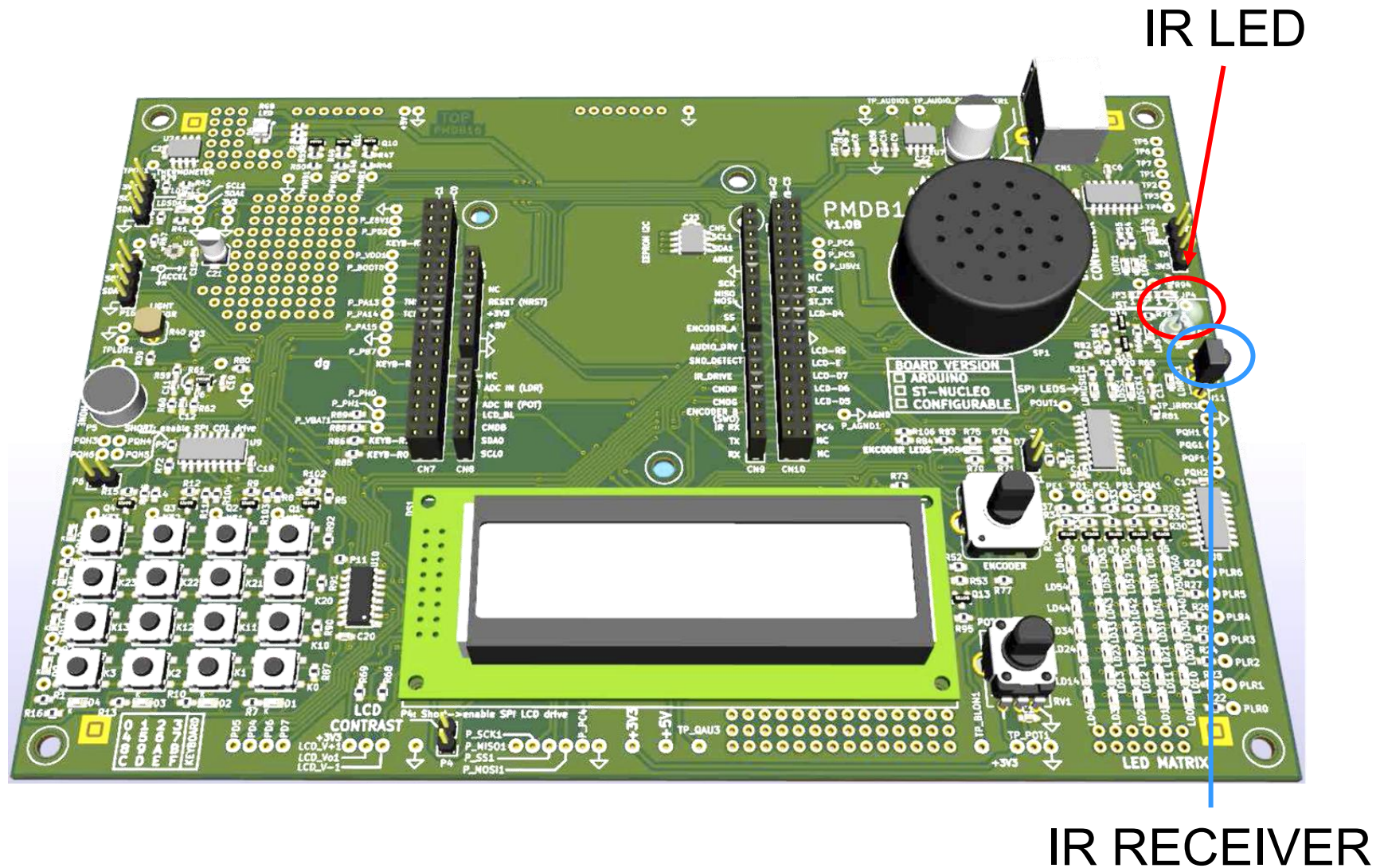
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# STM32 – IR communication

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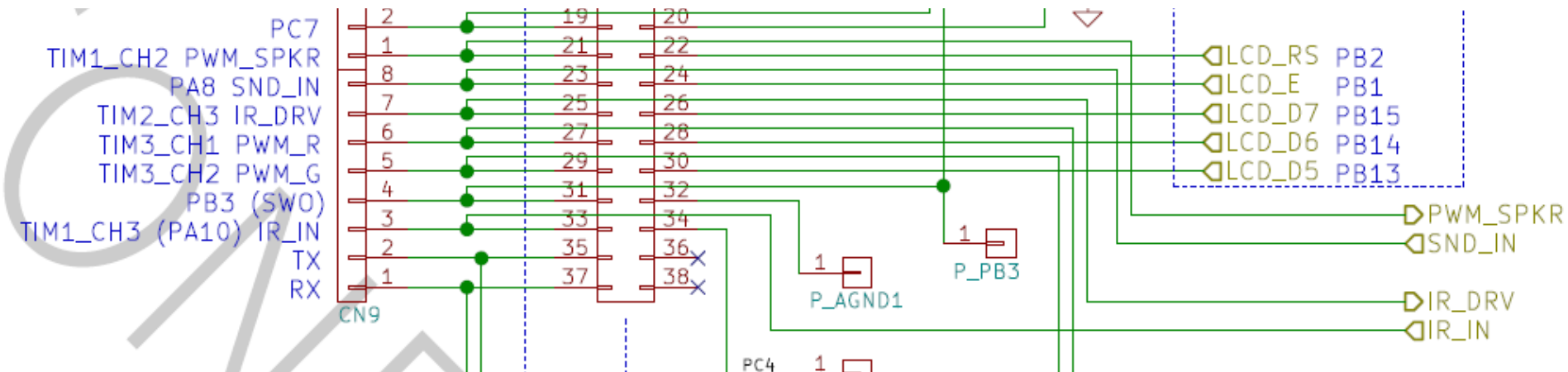




# IR TX / RX – PMDB16 details

In our PMDB16 board:

- IR LED (IR\_DRV) is connected to PB10 (TIM2\_CH3)
- IR receiver (IR\_IN) is connected to PA10 (TIM1\_CH3 / USART1\_RX)





# IR RX receiver – TSOP58238

Receiver consists of:

- Photodiode
- Amplifier with Automatic Gain Control (AGC)
- Bandpass filter and demodulator (38 kHz)
- Open-drain output with internal pull-up

Modulation allows to reject background light

Output goes **LOW** only when **MODULATED light is detected**

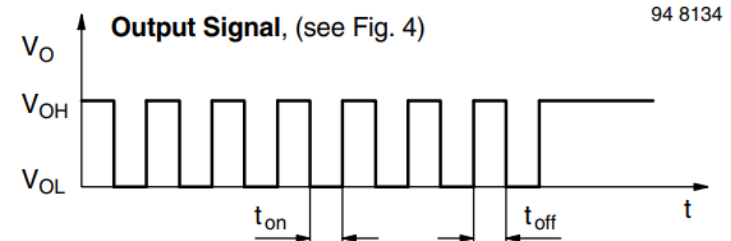
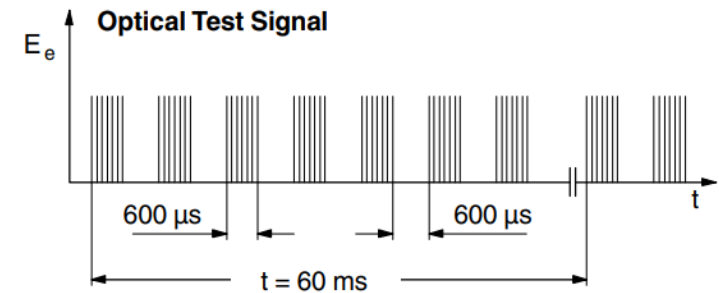
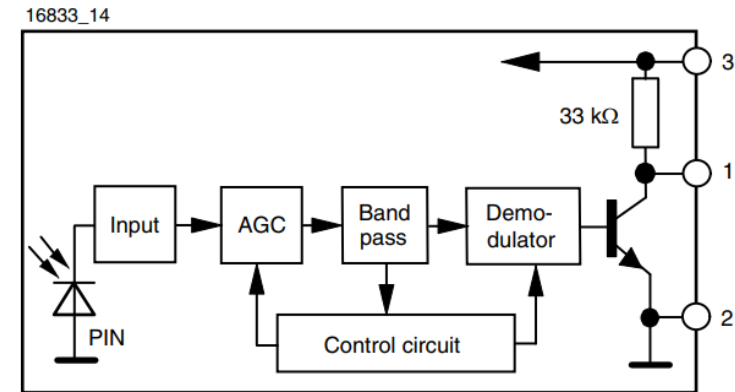


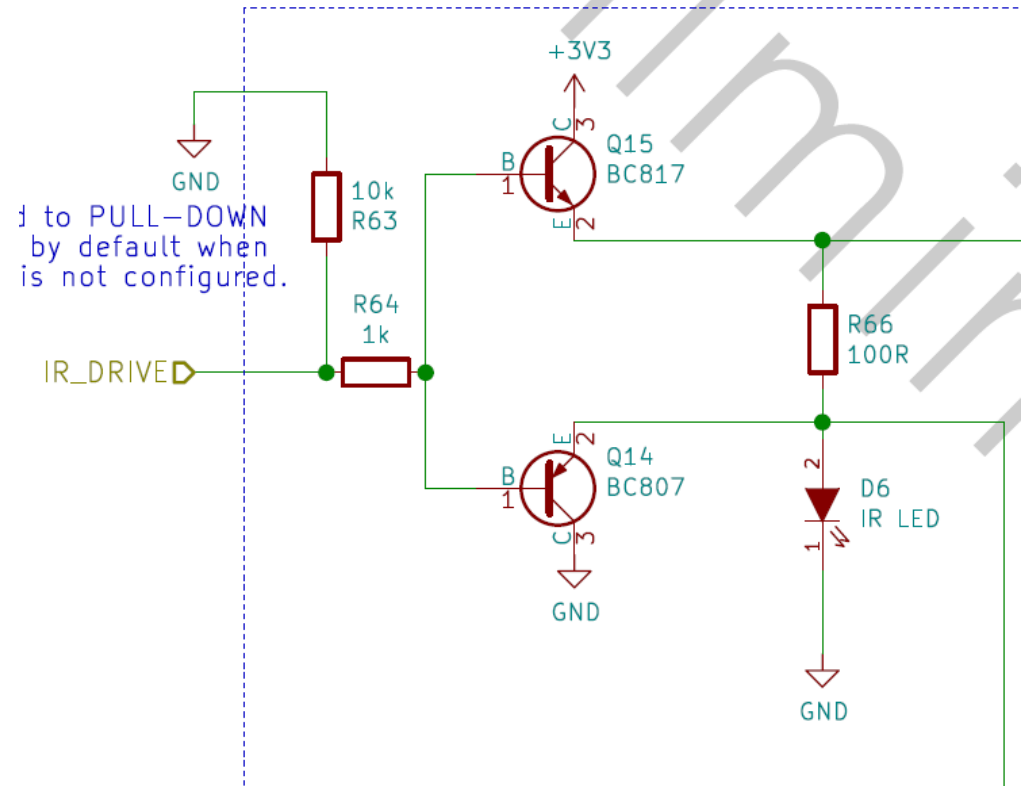
Fig. 3 - Output Function



## Transmitter:

- just a **simple LED**
- «Class B» amplifier to drive more current ( $\sim 20$  mA) than what the microcontroller pins allow (typ. 8 mA)

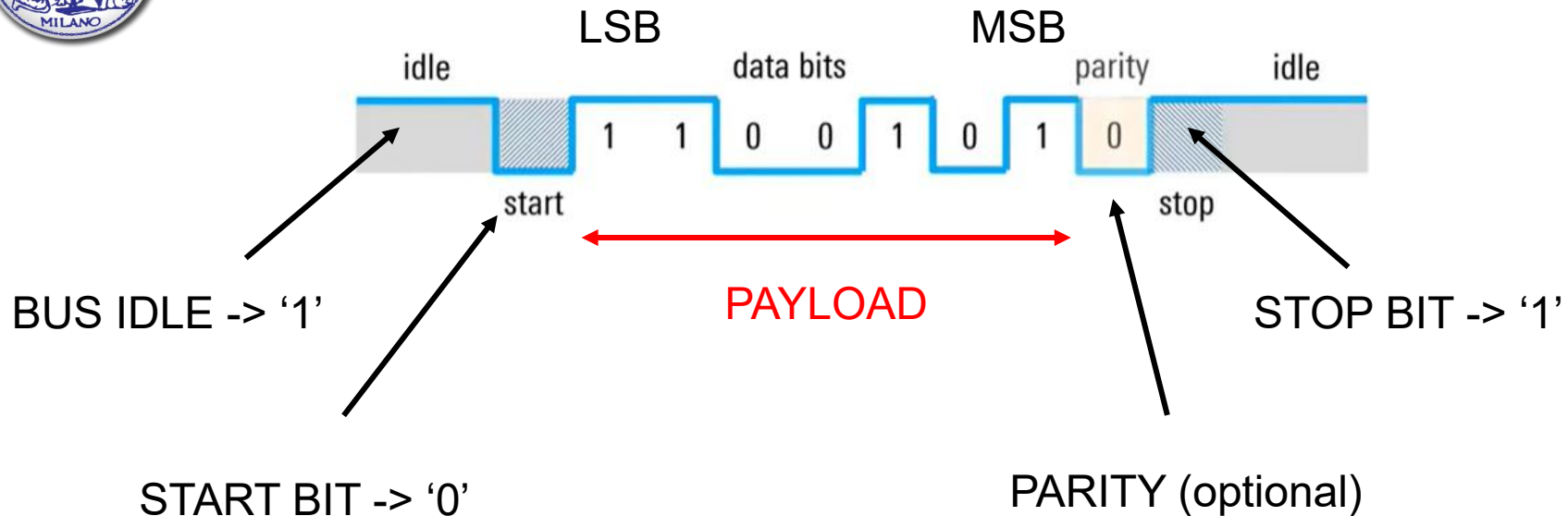
We can use **PWM** to drive the LED at 50% duty cycle, 38 kHz, to stimulate the receiver







# UART protocol



This diagram is what we want to get at the receiver side:

- To transmit (receive) a '1' → LED stays OFF
- To transmit (receive) a '0' → LED pulses at 38 kHz
- Bit time =  $1/(\text{Baud rate})$   
*Suggested baud rate  $\leq 2400$  bps*



The group should split into  
2 sub-groups:

## **Transmit and Receive**

Objective of this project is to  
prepare the two boards to allow  
wireless communication between  
them



# Transmit board – project 1

## Step 1:

Create a function that **sends a byte via «Infrared UART»** at 2400 bps baud rate:

- Set up a timer (*TIMx*) to provide an interrupt every  $1/2400$  s
- When the function gets called, enable *TIMx* in interrupt mode
- Send the start bit by enabling TIM2\_CH3 PWM at 38 kHz, until the *TIMx* interrupt
- Send the 8 bits, one at a time, enabling / disabling the PWM according to the bit to send
- Send the stop bit by disabling PWM

## Step 2:

Create a function to **send strings** exploiting what made in step 1.





# Receive board – project 1

## Step 1:

Receive data via UART, using USART2 in **interrupt mode**.

Suggestion: receive 1 byte in interrupt mode. When byte is received, immediately restart receive mode, and store previously received byte.

### React to received byte:

- Turn on green LED if you receive a '1'
- Turn off if you receive a '0'

You can debug by sending data via putty (just type in the window to send data).

## Step 2:

Switch to receiving from the IR USART1, at 2400 bps baud rate.

Send the byte received with USART1 to the PC via USART2, using both in interrupt mode. **Help the other half group debug their sending code.**



# Transmit - Project 2 - Objective

Objective

**Communicate between boards**



# Project objective

Objective of the project for the Transmit board is:

- Scan the **pushbutton** matrix (as in M13 project 1b) and **send** the data corresponding to the pressed button via IR UART

Objective of the project for the Receive board is:

- **Receive** the IR UART data and **display** the corresponding character on the **LED matrix** (as in M11 project 1b)

OR

Just play around and make the boards do whatever you like, as long as wireless communication via IR is involved



## Objective

**Integrate the previous project  
in the same board**